

Introduction to Mathematics and Optimisation

Exercises Week 3

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1 Exercise 1.13

An enemy transmits an encrypted version of a secret code (x_1, x_2, x_3, x_4) given by the following system of linear equations:

$$y_1 = 2x_1 + x_2 + 3x_3 + 4x_4$$

$$y_2 = x_1 + 2x_2 + 3x_3 + 4x_4$$

$$y_3 = 3x_1 + 3x_2 + x_3 + x_4$$

$$y_4 = 4x_1 + 4x_2 + 2x_3 + 3x_4$$

The encrypted message received is $(y_1, y_2, y_3, y_4) = (15, 16, 12, 20)$. The goal is to find the original secret code (x_1, x_2, x_3, x_4) .

1.1 Solution

We can take the system of equations and put it to matrix like so

$$\left(\begin{array}{cccc|c} 2 & 1 & 3 & 4 & 15 \\ 1 & 2 & 3 & 4 & 16 \\ 3 & 3 & 1 & 1 & 12 \\ 4 & 4 & 2 & 3 & 20 \end{array} \right).$$

Now we can perform these row operations to get the solution for the system (to save space I'm doing all divisions at the end):

$$\xrightarrow{\text{swap } R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 3 & 4 & | & 16 \\ 2 & 1 & 3 & 4 & | & 15 \\ 3 & 3 & 1 & 1 & | & 12 \\ 4 & 4 & 2 & 3 & | & 20 \end{pmatrix}$$

$$\xrightarrow{\begin{matrix} R_2 := R_2 - 2R_1 \\ R_3 := R_3 - 3R_1 \\ R_4 := R_4 - 4R_1 \end{matrix}} \begin{pmatrix} 1 & 2 & 3 & 4 & | & 16 \\ 0 & -3 & -3 & -4 & | & -17 \\ 0 & -3 & -8 & -11 & | & -36 \\ 0 & -4 & -10 & -13 & | & -44 \end{pmatrix}$$

$$\xrightarrow{\begin{matrix} R_3 := R_3 - R_2 \\ R_4 := 3R_4 - 4R_2 \end{matrix}} \begin{pmatrix} 1 & 2 & 3 & 4 & | & 16 \\ 0 & -3 & -3 & -4 & | & -17 \\ 0 & 0 & -5 & -7 & | & -19 \\ 0 & 0 & -18 & -23 & | & -64 \end{pmatrix}$$

$$\xrightarrow{R_4 := (-5)R_4 + 18R_3} \begin{pmatrix} 1 & 2 & 3 & 4 & | & 16 \\ 0 & -3 & -3 & -4 & | & -17 \\ 0 & 0 & -5 & -7 & | & -19 \\ 0 & 0 & 0 & -11 & | & -22 \end{pmatrix}$$

$$\xrightarrow{\begin{matrix} R_1 := 11R_1 + 4R_4 \\ R_2 := 11R_2 - 4R_4 \\ R_3 := 11R_3 - 7R_4 \end{matrix}} \begin{pmatrix} 11 & 22 & 33 & 0 & | & 88 \\ 0 & -33 & -33 & 0 & | & -99 \\ 0 & 0 & -55 & 0 & | & -55 \\ 0 & 0 & 0 & -11 & | & -22 \end{pmatrix}$$

$$\xrightarrow{\begin{matrix} R_1 := 55R_1 + 33R_3 \\ R_2 := 55R_2 - 33R_3 \end{matrix}} \begin{pmatrix} 605 & 1210 & 0 & 0 & | & 3025 \\ 0 & -1815 & 0 & 0 & | & -3630 \\ 0 & 0 & -55 & 0 & | & -55 \\ 0 & 0 & 0 & -11 & | & -22 \end{pmatrix}$$

$$\xrightarrow{R_1 := 1815R_1 + 1210R_2} \begin{pmatrix} 1098075 & 0 & 0 & 0 & | & 1098075 \\ 0 & -1815 & 0 & 0 & | & -3630 \\ 0 & 0 & -55 & 0 & | & -55 \\ 0 & 0 & 0 & -11 & | & -22 \end{pmatrix}$$

$$\xrightarrow{\begin{matrix} R_1 := \frac{1}{1098075} R_1 \\ R_2 := \frac{1}{-1815} R_2 \\ R_3 := \frac{1}{-55} R_3 \\ R_4 := \frac{1}{-11} R_4 \end{matrix}} \begin{pmatrix} 1 & 0 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & 0 & | & 2 \\ 0 & 0 & 1 & 0 & | & 1 \\ 0 & 0 & 0 & 1 & | & 2 \end{pmatrix}$$

From this we can read the solution

$$\boxed{x_1 = 1, \quad x_2 = 2, \quad x_3 = 1, \quad x_4 = 2}$$

1.2 Extra Credit

Suppose that you only know that the encryption scheme is

$$y_1 = a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4$$

$$y_2 = a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4$$

$$y_3 = a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4$$

$$y_4 = a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4$$

and that you have no knowledge of the numbers $a_{11}, \dots, a_{44}$. How many transmissions do you need to know at the minimum to find these encryption numbers?

1.2.1 Solution

We have 4 different unknowns on each row so we need 4 linear equations to solve these. In these equations the x_i values have to be linearly independent. So we need at minimum $\boxed{4}$ transmissions.

2 Exercise 2.35

Compute $a_0, a_1, a_2, a_3 \in \mathbb{R}$ so that

$$f(-2) = -1,$$

$$f(-1) = 1,$$

$$f(1) = 1,$$

$$f(2) = 1,$$

where

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

2.1 Solution

Substituting the given points to the given polynomial gives us this system of linear equations:

$$a_0 - 2a_1 + 4a_2 - 8a_3 = -1$$

$$a_0 - a_1 + a_2 - a_3 = 1$$

$$a_0 + a_1 + a_2 + a_3 = 1$$

$$a_0 + 2a_1 + 4a_2 + 8a_3 = 1$$

And as in the previous exercise we can put it into matrix:

$$\left(\begin{array}{cccc|c} 1 & -2 & 4 & -8 & -1 \\ 1 & -1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 1 \end{array} \right).$$

Now we perform the following row operations:

$$\begin{array}{l} R_2 := R_2 - R_1 \\ R_3 := R_3 - R_1 \\ R_4 := R_4 - R_1 \end{array} \longrightarrow \left(\begin{array}{cccc|c} 1 & -2 & 4 & -8 & -1 \\ 0 & 1 & -3 & 7 & 2 \\ 0 & 3 & -3 & 9 & 2 \\ 0 & 4 & 0 & 16 & 2 \end{array} \right)$$

$$\begin{array}{l} R_3 := R_3 - 3R_2 \\ R_4 := R_4 - 4R_2 \end{array} \longrightarrow \left(\begin{array}{cccc|c} 1 & -2 & 4 & -8 & -1 \\ 0 & 1 & -3 & 7 & 2 \\ 0 & 0 & 6 & -12 & -4 \\ 0 & 0 & 12 & -12 & -6 \end{array} \right)$$

$$R_4 := R_4 - 2R_3 \longrightarrow \left(\begin{array}{cccc|c} 1 & -2 & 4 & -8 & -1 \\ 0 & 1 & -3 & 7 & 2 \\ 0 & 0 & 6 & -12 & -4 \\ 0 & 0 & 0 & 12 & 2 \end{array} \right)$$

$$\begin{array}{l} R_3 := \frac{1}{6}R_3 \\ R_4 := \frac{1}{12}R_4 \end{array} \longrightarrow \left(\begin{array}{cccc|c} 1 & -2 & 4 & -8 & -1 \\ 0 & 1 & -3 & 7 & 2 \\ 0 & 0 & 1 & -2 & -\frac{2}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{6} \end{array} \right)$$

$$\begin{array}{l} R_1 := R_1 + 8R_4 \\ R_2 := R_2 - 7R_4 \\ R_3 := R_3 + 2R_4 \end{array} \longrightarrow \left(\begin{array}{cccc|c} 1 & -2 & 4 & 0 & \frac{1}{3} \\ 0 & 1 & -3 & 0 & \frac{5}{6} \\ 0 & 0 & 1 & 0 & -\frac{1}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{6} \end{array} \right)$$

$$\begin{array}{l} R_1 := R_1 - 4R_3 \\ R_2 := R_2 + 3R_3 \end{array} \longrightarrow \left(\begin{array}{cccc|c} 1 & -2 & 0 & 0 & \frac{5}{3} \\ 0 & 1 & 0 & 0 & -\frac{1}{6} \\ 0 & 0 & 1 & 0 & -\frac{1}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{6} \end{array} \right)$$

$$R_1 := R_1 + 2R_2 \longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{4}{3} \\ 0 & 1 & 0 & 0 & -\frac{1}{6} \\ 0 & 0 & 1 & 0 & -\frac{1}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{6} \end{array} \right)$$

Hence

$$a_0 = \frac{4}{3}, \quad a_1 = -\frac{1}{6}, \quad a_2 = -\frac{1}{3}, \quad a_3 = \frac{1}{6}$$

and the polynomial is

$$f(x) = \frac{4}{3} - \frac{1}{6}x - \frac{1}{3}x^2 + \frac{1}{6}x^3.$$